Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

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Abstract

Text watermarks usually hide a secret in token choices and later test those tokens without rerunning the generator. We ask whether a masked diffusion language model (DLM) can instead help verify its own output. Hiding selected parts of the context changes the probability of an emitted token in a reproducible, checkpoint-specific direction. Shibboleth uses that model response as the watermark carrier.

The method first finds positions where plausible candidates exhibit both response directions. Only after a position is fixed does the key choose which direction to favour. The generator then resamples from its current distribution without changing the decoding schedule. To verify a finished document, Shibboleth reconstructs the same masked contexts and counts how often the responses agree with the key. Position selection cannot observe the requested direction, so text produced independently of the key gives an exact binomial count.

At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA-8B-Instruct and 0.90 on Dream-7B-Instruct. The strongest token-local baseline reaches 0.93 on LLaDA but 0.57 or less on Dream when its LLaDA setting is transferred unchanged. At Shibboleth's main setting, the PPL ratio is $0.74\times$ on LLaDA and $1.09\times$ on Dream, and changes across the measured task scores are at most 0.06. This checkpoint binding has a cost: verification needs the same model and 144 full-sequence calls per 512-token document, and dispersed edits reduce detection power.

1 Introduction

When language-model text resembles human writing, its origin can be difficult to establish in settings such as disinformation analysis, phishing investigation, and academic-integrity review [28]. A post-hoc detector learns recurring output patterns; a watermark instead changes generation under a secret key and lets the key holder test a later document [21, 33]. We focus on watermarking because it can offer a stated finite-sample false-positive rate, rather than relying on a classifier whose behaviour may change with the decoder [10, 19, 23, 44].

Nearly all language-model watermarks place the secret in token identities: the key biases a vocabulary subset or couples randomness to each draw. Their detector therefore needs only the text and key (Figure 1a) [21, 24]. This design is inexpensive, but it must coexist with the decoder. For masked DLMs, that decoder may resolve many positions in parallel and choose their order from confidence, so forcing a resolved prefix or steering the order can change the system being watermarked [17].

This tension suggests a different question: can the originating model, rather than the token alone, carry the signal? A masked DLM can answer a reproducible question about its own output. Take a completed sentence, hide one subset of its earlier context, and record the probability of an emitted token; then hide a second subset and record it again. Some candidates become more likely and others less likely. The sign of that change is a *model response*. Because the verifier can recreate both hidings

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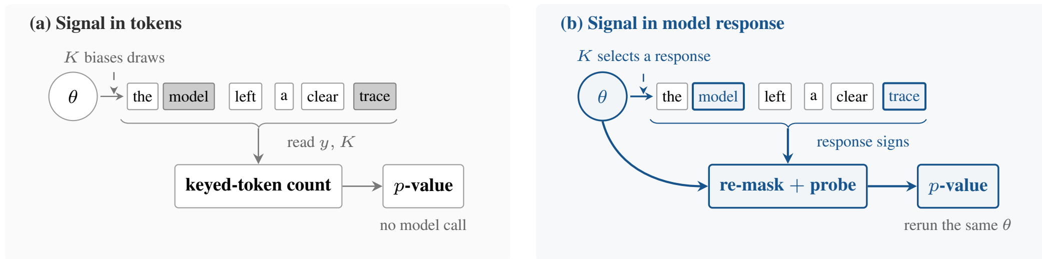

Figure 1: **Two places to encode a watermark.** Token-local methods alter sampled identities and verify from y, K alone. Shibboleth instead encodes which context response the checkpoint favours, so verification reruns the same θ.

from the finished text, this response can carry a keyed signal even when the token identity alone does not (Figure 1b).

Two obstacles separate this observation from a valid detector. First, verification must recreate the generation-time state even though the full continuation is now visible. Second, the method may seek positions where both response directions are plausible, but it must finish that search before the key reveals which direction counts as a match. Reversing this order would bias the selected positions toward the requested answer.

Shibboleth addresses both obstacles. For each active block, it re-masks that block and its suffix, compares pairs of hidden-prefix probes, and selects positions whose smaller directional probability total exceeds a fixed cutoff. The key then supplies the requested direction. At a selected position, Shibboleth draws up to R candidates from the decoder's current distribution and commits the first match; all other positions follow the reference decoder. Verification reconstructs the probes, repeats the selection rule, and evaluates an exact binomial tail. The block partition, confidence order, and checkpoint probabilities remain those of the reference decoder.

Our contributions are:

- We introduce model-response watermarking: the signal is a checkpoint's reproducible reaction to hidden context rather than a keyed pattern in token identities.
- We give a replayable construction whose carrier positions are selected before the keyed orientation is consulted, yielding finite-sample false-positive control while preserving the reference decoding schedule (Sections 3.2–3.5).
- We evaluate two masked DLMs. At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA and 0.90 on Dream. Verification requires 144 full-sequence model calls, and detection power decreases under dispersed edits (Section 4).

2 Related Work

Masked diffusion language models. Masked DLMs generate text by repeatedly predicting unresolved positions and committing a subset of them. This extends discrete diffusion to text [3, 16, 29, 32, 37, 38] and now supports instruction-following models including LLaDA and Dream [25, 31, 42, 45]. Which positions commit, and in what order, affects both quality and speed [2, 6, 20, 26, 39, 40]. We therefore treat the deployed schedule as a constraint: Shibboleth observes the model's masked-token predictions but does not choose a new commit order.

Watermarks encoded in tokens. Autoregressive watermarks usually bias a keyed vocabulary subset [21] or couple each draw to keyed randomness [1, 13, 24]. Their detectors are cheap because a key–token calculation needs no model call, and later work characterises when those detectors are reliable or undetectable [10, 22, 44]. DLM methods adapt this idea by removing prefix dependence, averaging a red–green bias over unresolved context, sampling with keyed randomness, or encoding information in the denoising order [4, 9, 14, 17, 34, 41]. Our closest baseline, dgMARK, changes which mask resolves next and later reads that order from the text. Shibboleth instead keeps the order fixed and asks the checkpoint to reproduce its response.

Detectors that use a model. Model access can support richer tests than a key–token hash. Likelihood-ratio detectors compare watermarked and base conditionals [18]; access to the model can improve a Neyman–Pearson score over a key-only Gumbel statistic [12]; GaussMark verifies a keyed weight perturbation with a forward–backward pass [7]; and model-theft watermarks query a suspect system [43]. Outside watermarking, DetectGPT scores text using log probabilities from the model of interest and perturbed versions of the passage; replacing the source model with a surrogate weakens detection [30]. These model-assisted detectors are most natural when a provider or central verifier can afford model access. Shibboleth shares that operational setting but adds a keyed generation channel and a document-level test. It leaves the checkpoint unchanged and keys the contexts presented to it, so one deployed model can serve many keys.

Finite-sample detection and editing. Watermark tests increasingly replace asymptotic z-scores with exact tails, permutation tests, or corrected maxima over windows [12, 22, 24]. After editing, a statistic that concentrates on surviving regions can outperform a document-wide sum [27]. Shibboleth uses an exact binomial tail for the unedited detector and compares document-wide, block-local, and windowed aggregation under editing in Section 4.2.

3 Shibboleth: Watermarking by Model Response

3.1 Masked Decoding

A masked DLM predicts many unresolved positions at once rather than exposing a unique "next token." Given a sequence containing masks, it returns a distribution at every unresolved position. Let x be the prompt, $y = (y_1, \ldots, y_N)$ the continuation, and I the positions already revealed. At any masked position $i \notin I$, the model provides $p_\theta(y_i \mid y_I, x)$. One forward pass over the full sequence provides these distributions for every unresolved position; we count this as one full-sequence model call.

We study the blockwise schedule used by LLaDA [2, 31]. Blocks finish from left to right, while a confidence rule decides which masks inside the active block commit at each step. We write B_b for the active block and F_b for the completed prefix before it. Shibboleth keeps both the block order and this confidence rule.

3.2 Problem Setup

We target keyed provenance detection: distinguish text generated with a secret document key from text produced independently of that key, at a false-positive rate fixed in advance. The watermark may change which token is sampled, but should not replace the model's decoder or its schedule.

What the two parties retain. The embedder has the model, prompt, secret key K, and target bits. The verifier later has the completed sequence, the same checkpoint, K, and public parameters—but no generation trace, stored logits, random seed, or record of rejected samples. A nonce derives a fresh document key $K_d = \mathrm{HMAC}(K, \mathrm{nonce}_d)$ [5]. Separate hash domains use K_d for probe construction, carrier assignment, and the requested response directions.

Guarantee and scope. For any text produced independently of K_d, including output from another sampler, a detector run at level α fires with probability at most α. This guarantee is finite-sample and permits the method to choose a data-dependent number of carriers. It does not promise detection after editing, key disclosure, adaptive detector queries [19], or replay with a numerically different checkpoint.

Why write during generation? Editing only after completion offers too little freedom at most positions. Across 36 post-hoc substitution settings, the median selected position has only one candidate within a 3-nat likelihood budget; the best setting below a $1.35\times$ PPL budget reaches 0.10 TPR at 1% FPR. Shibboleth therefore writes while the model's live distribution is still available.

The design follows three requirements:

1. **Replayability.** From the completed sequence, the verifier must reconstruct the same masked context that the embedder used.

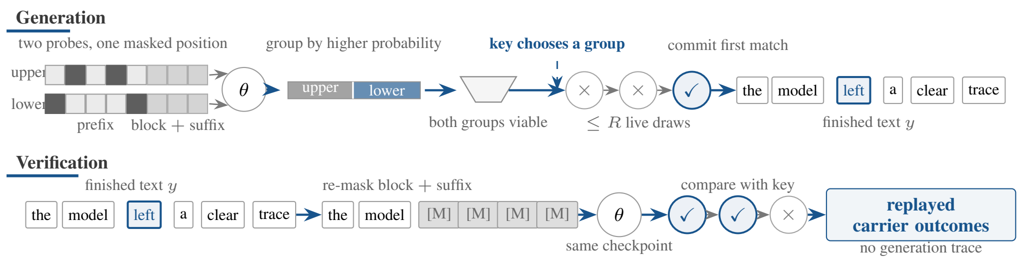

Figure 2: **Probe pair and replay.** Two hidden-prefix probes split candidate tokens by which probe assigns more probability. Both groups must be viable before the key chooses one; verification recreates the same probes from the finished text.

2. **Blind selection.** Carrier positions must be fixed before the key reveals which response direction counts as a match.
3. **Schedule preservation.** Blocks and within-block commit order must remain those of the reference decoder.

Replayability makes a generation-time response measurable later. Blind selection keeps adaptive carrier discovery from biasing the detector, while schedule preservation limits changes to the deployed generator. The next three subsections implement these requirements.

3.3 Probing the Model and Selecting Carriers

Figure 2 first locates the main operations within one generation-and-verification block. For each unfinished block, Shibboleth asks the model the same question under several masked-prefix probes: how likely is each candidate token when this part of the earlier context is missing? Comparing two probes splits candidate tokens by which probe assigns them more probability. The position can carry a bit only when the live decoder assigns sufficient probability to both groups; only then does the key choose which group the emitted token should favour.

Formally, Shibboleth constructs L pseudorandom probe patterns $D_{b,\ell} \subset F_b$, where each pattern specifies prefix positions to hide. It also masks the current block B_b and every later position. Let $\lambda_{i,\ell}(v)$ denote the resulting log-probability of vocabulary item v at position i under pattern ℓ. The block and suffix are hidden because neither was available when the probe was created. The verifier can therefore reconstruct the same state from the completed sequence instead of seeing later tokens that would alter the probabilities. Probing a block costs $L+1$ full-sequence model calls: L masked-prefix probes and one evaluation with the full prefix visible.

For a pair (ℓ, ℓ'), define the response $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} be the block-masked distribution with no hidden prefix. The aggregate probability assigned to either response direction, and a symmetric score for the pair, are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z)\mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \qquad S_i^{\ell,\ell'} = 2\min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \tag{1}$$

The score is twice the smaller total. It approaches one when the live distribution is divided evenly between the two response groups and approaches zero when one group has little probability. At each position, Shibboleth keeps the pair with the largest S_i and selects the position when $S_i > s_{\min}$, subject to an optional per-block cap. Choosing the best pair raises the measured mean score from 0.28 for a fixed near/far pair to 0.45. Figure 3 shows where this criterion retains or skips positions in one saved LLaDA output.

Crucially, swapping ℓ and ℓ' leaves S_i unchanged while reversing every response sign. Carrier selection therefore cannot favour the direction later requested by the key.

After the carrier is fixed, the key assigns one group to the target bit. Formally, a separate hash domain supplies $\epsilon_i \in \{-1, +1\}$, while $w_i \in \{-1, +1\}$ represents the provenance or payload bit. Together they decide which response direction counts as a match:

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \tag{2}$$

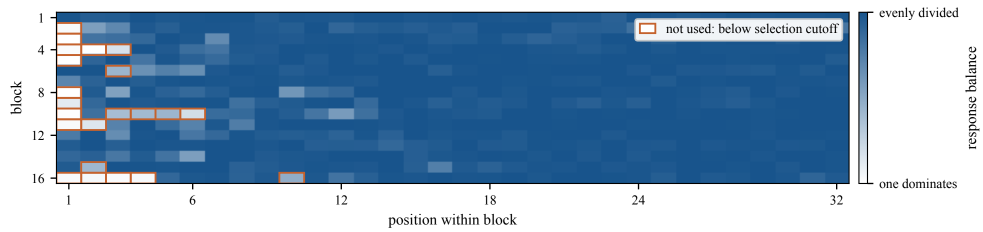

Figure 3: **Carrier availability in one generated document.** Each cell is one token position. Darker blue indicates greater balance between the response groups in Figure 2; orange outlines mark positions below the cutoff, where no bit is written.

If $g_i(z) = 0$, a second domain-separated bit assigns the outcome. Such a tie cannot be steered, but it remains an independent fair outcome for detection.

3.4 Embedding Without Changing the Schedule

The reference decoder still chooses the next position and supplies its live distribution p_i. Unselected positions are sampled once as usual. At a selected carrier, Shibboleth accepts a candidate only if it has the requested response and is not too far below the most likely token:

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \;\text{ and }\; p_i(z) \geq \kappa \max_v p_i(v) \right\}. \tag{3}$$

It draws from the same p_i up to R times and commits the first member of A_i. If every proposal misses, it takes one fresh unchecked draw. All retries reuse p_i, so they add no model call and do not alter the commit order.

Let $a_i = p_i(A_i)$ be the probability of satisfying both conditions, and let $q_i = \sum_z p_i(z) m_i(z)$ be the chance that an unchecked draw scores as a match. The committed token comes from $p_i(\cdot \mid A_i)$ with probability $1 - (1 - a_i)^R$ and from p_i otherwise. Its match probability is therefore

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \tag{4}$$

With no plausibility floor and no ties, $a_i = q_i$, giving $1 - (1 - q_i)^{R+1}$. Equation (4) separates the two controls: R determines how many chances the sampler has to find an acceptable token, while κ limits how unlikely that token may be relative to the live argmax. Larger R can strengthen detection but changes the sampling distribution; κ does not constrain the unchecked fallback or guarantee semantic quality, which we measure separately.

3.5 Watermark Detection

The verifier re-masks each completed block and suffix, reconstructs the probe pairs and carriers, and evaluates $m_i(y_i)$. It needs no rejected samples or generation trace. For provenance detection, the selected carriers form a pool P with $w_i = +1$; a payload implementation would keep a separate pool for each message bit. The document statistic and its one-sided p-value are

$$T = \sum_{i \in P} m_i, \qquad p_{\text{det}} = \Pr\left[\text{Binomial}(|P|, \tfrac{1}{2}) \geq T\right]. \tag{5}$$

Figure 4 follows a five-carrier example from replayed outcomes to the exact upper-tail probability.

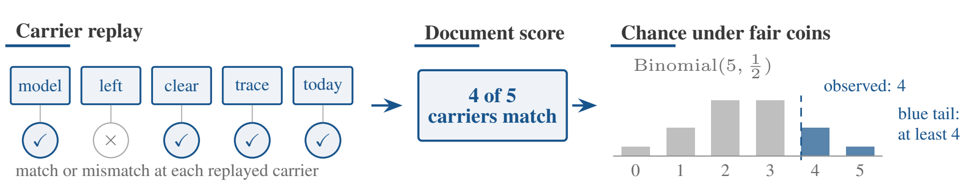

Figure 4: **Document-level detection.** The verifier replays each carrier and records a match. Here four of five match; the blue bars give the exact chance of observing at least four matches under fair coins.

Method	Setting	PPL ratio ↓	TPR at FPR ↑ 1%	TPR at FPR ↑ 0.1%	Full-sequence model calls ↓ Generation	Full-sequence model calls ↓ Verification
LLaDA-8B-Instruct						
KGW [21]	$\delta = 1$	1.22×	0.90	0.83	512	**0**
dgMARK [17]	$k = 1$	1.45×	0.83	0.80	512	**0**
dgMARK [17]	+3-beam	1.21×	**0.93**	**0.93**	∼2048	**0**
Shibboleth	$R = 1$	0.97×	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.87×	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$	**0.74**×	0.80	0.60	656	144
Dream-7B-Instruct						
KGW [21]	$\delta = 1$	1.84×	0.57	0.37	512	**0**
dgMARK [17]	$k = 1$	1.75×	0.37	0.27	512	**0**
dgMARK [17]	+3-beam	**0.71**×	0.30	0.30	∼2048	**0**
Shibboleth	$R = 1$	1.34×	0.40	0.27	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.79×	0.70	0.67	656	144
Shibboleth	$R = 16, \kappa = 0.05$	1.09×	**0.90**	**0.90**	656	144

Table 1: **Detectability and cost at 512 tokens.** PPL ratio is measured against each method's own unwatermarked run and should be interpreted within that method. Shibboleth uses $n = 30$ documents ($n = 20$ for $R = 16$); Dream PPL uses 8–13 successfully scored document pairs per setting. Settings chosen on LLaDA are transferred to Dream without retuning. One model call is one forward pass over the current full sequence; these counts measure algorithmic work, not runtime. δ is KGW's logit bias and k is dgMARK's candidate count.

Proposition 1 (Exact false-positive control)**.** Assume the hash domains behave as independent random functions. For any text written without the document key, condition on the selected carriers, probe pairs, responses, and target bits. The orientation bits remain independent fair coins. Hence the m_i are independent $\mathrm{Bernoulli}(\frac{1}{2})$ variables, $T \sim \mathrm{Binomial}(|P|, \frac{1}{2})$, and thresholding Eq. (5) at α gives false-positive rate at most α for every document length and carrier count.

Why adaptive carrier selection is allowed. The text and model may determine the number of carriers, the probe pair, and the response size. What matters is that each choice is fixed before the independent orientation bit is used. Flipping ϵ_i then flips the match whenever $g_i(y_i) \neq 0$; the tie rule supplies a fresh fair bit otherwise. The detector therefore needs neither an asymptotic approximation nor a calibration corpus. The guarantee still assumes text produced independently of the key and domain separation among the key's several roles.

4 Experiments

The evaluation asks four questions: detection strength, transfer across checkpoints, computational and quality cost, and behaviour under editing.

Models and data. We evaluate LLaDA-8B-Instruct [31] and Dream-7B-Instruct [42] in FP16 on one TITAN RTX. Detectability uses C4 `realnewslike` prompts [36], truncated to 300 characters, and 512-token continuations divided into 32-token blocks. We correct Dream's one-position logit offset in its wrapper, then use the same position convention for generation and replay.

Decoders and settings. Shibboleth and dgMARK use each model's confidence schedule with multinomial sampling at temperature 0.8; dgMARK additionally uses top-$k = 3$. KGW requires strictly left-to-right decoding. Shibboleth uses $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R, and optional plausibility floor κ. We evaluate provenance ($w_i = +1$), not payload recovery. Here one model call means one forward pass over the current full sequence. With 16 blocks and $L + 1 = 9$ probe evaluations, verification replays $16 \times 9 = 144$ calls; generation uses 512 ordinary decoding calls plus the same 144 probe calls, for 656 in total.

Metrics. Detection is TPR at the analytic FPR thresholds in Eq. (5). Perplexity is scored by GPT-2-large [35] as $\exp(\mathrm{mean}(\mathrm{NLL}_{\mathrm{wm}} - \mathrm{NLL}_{\mathrm{unwm}}))$. A PPL ratio therefore compares a method with its own unwatermarked decoder; it is not a semantic-quality score and is only indicative across different decoders. The baselines are KGW [21] and dgMARK [17].

Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct			
KGW [21]	0.530→0.535	0.640→0.740	0.329→0.293
dgMARK [17]	0.605→0.530	0.740→0.660	0.409→0.341
dgMARK +3-beam	0.605→0.530	0.740→0.660	0.409→0.280
Shibboleth	0.595→0.575	0.640→0.700	0.335→0.384
Dream-7B-Instruct			
KGW	0.640→0.625	0.700→0.700	0.341→0.421
dgMARK	0.740→0.570	0.840→0.560	0.567→0.421
dgMARK +3-beam	0.740→0.645	0.840→0.560	0.567→0.329
Shibboleth	0.695→0.710	0.660→0.680	0.427→0.439

Table 2: **Task accuracy before and after watermarking.** Each cell is unwatermarked → watermarked under that method's decoder. MMLU uses 200 questions, GSM8K 50 problems, and HumanEval 164 executed tasks. Approximate standard errors are 0.035, 0.07, and 0.037, respectively; — denotes an unmeasured pair. Shibboleth uses $R = 16, \kappa = 0.05$.

4.1 Main Results and Analysis

Table 1 shows the tradeoff: Shibboleth transfers strongly to Dream but replay is costly, while two cheaper baselines detect LLaDA output more often.

Retries trade sampling distortion for detection. On LLaDA, $R = 1$ reaches 0.43 TPR at 1% FPR. On 30 held-out prompts, $R = 2, 4, 8$ increase TPR to $0.73, 0.83, 0.87$, consistent with Eq. (4). Retries add no model evaluation, but without a floor the PPL ratio rises from $1.14\times$ at $R = 2$ to $1.61\times$ at $R = 8$. The floor removes low-probability guided draws: $(R, \kappa) = (8, 0.1)$ gives 0.70 TPR at $0.87\times$ PPL, and $(16, 0.05)$ gives 0.80 at $0.74\times$. Ratios below one mean only that GPT-2-large prefers the retained high-probability tokens.

The transferred setting behaves differently across checkpoints. We carry every LLaDA setting to Dream without retuning. KGW reaches 0.57 TPR at 1% FPR and produces 84% duplicate bigrams under forced left-to-right decoding. Direct and three-beam dgMARK reach 0.37 and 0.30 TPR. Shibboleth at $(16, 0.05)$ reaches 0.90 at both reported FPR levels. Its PPL ratio is $0.79\times$ at $(8, 0.1)$ and $1.09\times$ at $(16, 0.05)$, based on 8–13 scored pairs per setting. Thus detectability transfers in this experiment, whereas the setting with the lowest LLaDA PPL ratio does not retain that ratio on Dream.

Task accuracy exposes decoder-specific costs. Table 2 shows each method's own unwatermarked and watermarked scores side by side; Appendix Table 5 gives the full grid. Shibboleth changes by at most 0.06. On LLaDA HumanEval, dgMARK changes from 0.409 to 0.341 direct and 0.280 with three beams; on Dream, MMLU changes from 0.740 to 0.570, GSM8K from 0.840 to 0.560, and HumanEval from 0.567 to 0.329 with three beams. On 256-token LLaDA GSM8K answers, Shibboleth and direct dgMARK reach 0.12 TPR at 1% FPR, three-beam dgMARK 0.44, and KGW zero; 512-token rates therefore do not transfer to short answers.

Checkpoint binding costs model calls. On LLaDA, Shibboleth's $0.80/0.60$ TPR at $1\%/0.1\%$ FPR trails three-beam dgMARK's $0.93/0.93$ and KGW's $0.90/0.83$, while requiring 144 verification calls instead of zero. KGW's left-to-right decoder also repeats heavily: 37% of unwatermarked documents cross the threshold before bigram deduplication, and two of thirty cross 0.1% afterward. With 20–30 documents, TPR differences near 0.1 remain imprecise. Wall-clock, verification takes 26.8 s per document on one TITAN RTX (KGW 0.17 s, dgMARK under a millisecond); the cost is a dial rather than a fixed price, since $L = 4$ probe patterns cut the replay to 80 calls at 0.70 TPR at 1% FPR against 0.77 at $L = 8$, while $L = 2$ saves only 32 further calls and falls to 0.57 (Appendix Table 6).

4.2 Calibration and Robustness

Observed false-positive rates are below nominal levels. Across 20 keys and 120 unwatermarked LLaDA generations per key, mean FPR is $0.0121/0.0037/0.0000$ at nominal $0.10/0.05/0.01$; the largest per-key rates are $0.0583/0.0250/0.0000$. Every Dream cohort has zero false positives at 1% (Appendix Table 3).

Dispersed edits break prefix synchronisation. At $R = 1$, random 10% re-denoising or deletion reduces TPR from 0.35 to 0.05. At $R = 16$, $\kappa = 0.05$, random 5% and 10% edits leave 0.88 and 0.50 TPR, whereas a contiguous 10% or 30% span leaves 0.88 because earlier carriers survive. Blockwise Bonferroni and 128-token windowing reach only 0.38 and 0.44 under random 10% editing, below pooled counting at 0.50 (Appendix Table 4). Sparse survival crosses over only at extreme destruction: pooled counting holds 0.81 with half the document re-denoised in place, and at 75% pooled and Bonferroni tie at 0.50. The same attack barely moves the token-local detectors (KGW 0.90 to 0.85, dgMARK 0.85 to 0.80 at 1%, n=20): it is hard for prefix-synchronised replay specifically, not for watermarks generally.

5 Discussion

Deployment role. Shibboleth is most useful when the question concerns a registered model release — whether that checkpoint is consistent with a disputed document. Providers, centralised services, or auditors can retain the model without changing its production decoder. The method is a poor fit for browser-scale classification, unknown source models, short fragments, or heavily edited text. An inexpensive token-local detector can handle routine cases, with replay reserved for disputes that justify model-specific computation.

Reducing the verifier burden. Clients need not execute the model: a provider can replay a document on the registered checkpoint and return a signed report, or an independent auditor can use escrowed weights. Batching improves hardware utilisation, but each document induces different masked contexts, so replay cannot become a fixed lookup; a surrogate checkpoint or cached responses may reconstruct different carriers and would need their own false-positive analysis.

Trust boundary. A hosted verifier assumes that the service runs the registered checkpoint and receives the submitted document. A signed report records the service's claim; it does not prove correct execution or protect document privacy. Checkpoint and container hashes, append-only audit records, and independent execution can strengthen this arrangement; public proof of correct replay, attestation, service compromise, and adversarial requests are not evaluated here.

Interpreting a detection result. A low p-value has a narrow interpretation: under the registered key and replay procedure, the document contains more checkpoint-consistent carrier responses than expected by chance. It does not identify the person who requested the text, establish that every passage came from the model, or rule out editing; those questions require separate records. A report should therefore include the p-value, carrier count, key identifier, checkpoint hash, precision, probe construction, and threshold; the count separates a short, inconclusive document from checkpoint disagreement.

6 Conclusion

Shibboleth encodes a keyed signal in reproducible model responses to hidden context. Selecting a carrier with a symmetric score before revealing the keyed direction yields an exact binomial test even with a data-dependent carrier count. The method preserves the deployed schedule and reaches 0.80 TPR on LLaDA and 0.90 on Dream without retuning, but needs 144 same-checkpoint calls per 512-token document, so it complements cheaper token-local detectors, which perform better on LLaDA.

Limitations. The false-positive guarantee does not imply editing tolerance: random 10% re-denoising reduces TPR from 0.35 to 0.05 at $R = 1$ and from 0.88 to 0.50 at $R = 16$, while a contiguous span leaves earlier carriers intact. A PPL ratio below one does not imply better text, and the task cohorts are too small to interpret modest changes. Replay requires the same checkpoint and precision; evaluation covers two masked DLMs, one corpus, 16–50 documents per cohort, and provenance only.

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A Additional Tables

Table 3 reports the measured per-key false-positive rate of the deployed decision rule. Table 4 summarises the post-editing evaluation under the deployed pooled detector. Table 5 expands Table 2 into the full task-accuracy grid, including the greedy-decoding cells. Table 6 shows how detection and verification cost trade off as the number of probe patterns varies.

Nominal α	Mean FPR	Highest per-key FPR
0.10	0.0121	0.0583
0.05	0.0037	0.0250
0.01	0.0000	0.0000

Table 3: **Observed false-positive rates across keys.** The evaluation uses 20 keys and 120 unwatermarked generations per key under the deployed decision rule.

		TPR at FPR ↑	
Attack	Strength	1%	0.1%
Configuration A: $R = 1$, 512 tokens ($n = 20$)			
None	—	0.35	—
Re-denoising (same model, argmax)	10% of positions	0.05	—
Deletion	10% of positions	0.05	—
Insertion	10% of positions	—	—
Substitution (random)	10% of positions	—	—
Copy–paste	25% kept, 1 segment	—	—
Paraphrase (DIPPER) [23]	lex 20, order 0	—	—
Configuration B: $R = 16, \kappa = 0.05$, 1024 tokens ($n = 16$)			
None	—	0.88	0.88
Re-denoising (random positions)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38
Re-denoising (contiguous span)	10% of positions	0.88	0.88
	30% of positions	0.88	0.88
	50% of positions	0.81	0.69
	75% of positions	0.50	0.31

Table 4: **Detection after editing.** The pooled detector uses the same analytic thresholds as Table 1. Random edits disrupt later prefixes; a contiguous span preserves all earlier carriers. Because the cohorts are small, one document changes TPR by 0.05 when $n = 20$ and about 0.06 when $n = 16$. An em dash denotes an unmeasured setting.

		Greedy decoding			Multinomial decoding		
Model	Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct	Non-watermarked	0.605	0.740	0.409	0.595	0.640	0.335
	KGW [21]	—	—	—	0.535	0.740	0.293
	dgMARK [17]	—	—	—	0.530	0.660	0.341
	dgMARK +3-beam	—	—	—	0.530	0.660	0.280
	Shibboleth	—	—	—	0.575	0.700	0.384
Dream-7B-Instruct	Non-watermarked	0.740	0.840	0.567	0.695	0.660	0.427
	KGW	—	—	—	0.625	0.700	0.421
	dgMARK	—	—	—	0.570	0.560	0.421
	dgMARK +3-beam	—	—	—	0.645	0.560	0.329
	Shibboleth	—	—	—	0.710	0.680	0.439

Table 5: **Text generation quality, full grid.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16, \kappa = 0.05$; 50 problems per GSM8K cell, so differences in individual scores should be interpreted cautiously. Each watermark decodes under its own regime—KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell)—so rows are read against their own controls, not across. An em dash denotes an unmeasured setting.

L	Verify calls ↓	TPR at FPR ↑			Sync rate
		5%	1%	0.1%	
2	48	0.73	0.57	0.50	0.629
4	80	0.77	0.70	0.67	0.650
8	144	0.77	0.77	0.73	0.658

Table 6: **Fewer probe patterns make verification cheaper.** Each row halves the number of probe patterns L; verification replays $16(L+1)$ model calls. Detection is measured at $R = 16, \kappa = 0.05$ on LLaDA, 512 tokens, $n = 30$ fresh prompts. Going from $L = 8$ to $L = 4$ keeps most of the detection at a little over half the cost; going to $L = 2$ saves little more and loses about a tenth of TPR at every operating point.

B Per-Document Carrier Statistics

Figure 5 reports per-document statistics from a verifier replay over the saved 512-token arms. Carrier counts average 271 (LLaDA control), 192 (LLaDA R=8, κ=0.1), 214 (Dream control) and 208 (Dream R=16, κ=0.05), with wide spread ($\approx$ 30 to 500) driven by early termination, since EOS padding admits few carriers. The per-document Wilson intervals separate documents whose evidence is individually decisive from the minority that overlap the null.

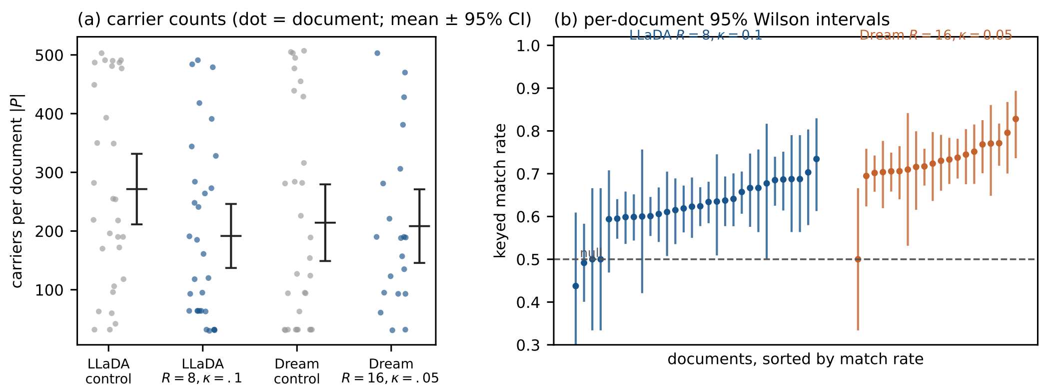

Figure 5: **Per-document carrier counts and detection intervals.** (a) Carrier pool size $|P|$ per document for the control and watermarked arms of both models (dot = document; whisker = mean $\pm$ 95% CI). Counts range from about 30 to 500: short, EOS-padded documents admit few carriers. (b) Per-document 95% Wilson intervals on the keyed match rate for the watermarked arms, sorted; documents whose interval clears the 0.5 null carry individually decisive evidence, and the weakest documents overlap it.